

On the Predictability of Parking Preferences

Takamasa Higuchi and Kentaro Oguchi
Toyota Motor North America R&D
Mountain View, CA, USA

ABSTRACT

The recent advances in vehicular crowd sensing technology have paved the way for city-scale monitoring of parking spot availability. It enables intelligent parking navigation with the spot-level granularity, guiding drivers to the most convenient available spot within a busy parking lot. Since people would have various preferences on parking spot selection, personalization is a key to improve user experience in such parking navigation systems. In this paper, we investigate the feasibility of predicting a parking spot that a user would prefer out of a given set of available candidates.

KEYWORDS

Intelligent Parking, Personalization

ACM Reference Format:

Takamasa Higuchi and Kentaro Oguchi. 2021. On the Predictability of Parking Preferences. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*, September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3460418.3479273>

1 INTRODUCTION

Despite the evolution of vehicle navigation services, the problems associated with parking search are still largely unaddressed. People in the U.S. spend tens of hours a year for parking search even after arriving in their destinations [1]. The parking seekers circling around the city in turn increase road congestion and CO2 emissions [4]. The recent advances in intelligent parking solutions may provide potential remedies for these societal issues. Some local governments conducted pilot studies, deploying tens of thousands of roadway occupancy sensors that monitor live availability of parking spots [4]. The research community has also delved into a crowd-sourcing approach to parking availability sensing [2, 3]. They use vehicle on-board sensors to measure availability of the surrounding parking spots while traveling on a road, and aggregate information from a large fleet of vehicles to enable wide-scale availability monitoring without dedicated sensor infrastructure. These trends have paved the way for spot-level parking navigation.

People would have various preferences on parking spot selection. Some people may prefer parking spots that are as close as possible to the destination (e.g., entrance of a retail store), while some others may consider ease of parking as the most important factor. Personalization would be a key to guide users to a preferred

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8461-2/21/09.

<https://doi.org/10.1145/3460418.3479273>

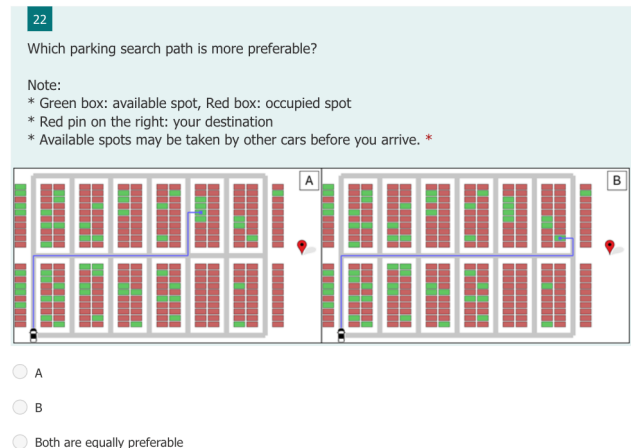


Figure 1: Example of user study questions

spot that meets their own intuition and priority. In this paper, we investigate the feasibility of building a decision making model that accurately mimics users' parking spot selection.

2 USER STUDY

We used Amazon Mechanical Turk to conduct a preliminary user study. Each participant is shown pairs of hypothetical parking lot maps as in Figure 1, which indicate availability of each parking spot. The green rectangles signify available spots while red spots being occupied. Each pair of maps have the same parking availability conditions, while showing different candidate parking search paths ego vehicle may take. Once a vehicle is parked in the target parking spot, passengers will walk to the destination, indicated by a red pin icon. The participants are asked to choose the preferable parking search path according to their own preferences. The participants may also give a neutral response if both options are equally preferable. They perform the pair-wise comparisons over 50 pairs of scenarios with different parking availability conditions. We collected responses from 89 participants. Five of them were filtered out as "poor-quality" responses because the response time was less than a half of the average time over all the participants. This constitutes a data set of 4,200 responses from 84 participants.

3 PREFERENCE MODELING FRAMEWORK

Our goal is to build parking preference model(s) that predict a parking spot a user chooses under a given availability condition. We assume that the system may use the history of parking search paths (e.g., from GPS trajectories) and the live availability of parking spots, measured by crowdsourced availability sensing [2, 3].

We analyze each parking availability map to extract the following four features that represent characteristics of a given candidate parking search path:

- *Distance to walk* (s_{walk}): distance from the target parking spot to user's destination (in meters).
- *Time to drive* (s_{drive}): expected time to move from the vehicle's current location to the target parking spot (in seconds), which is calculated based on the distance to drive and the speed limit of each road segment.
- *Ease of parking* (s_{ease}): 0.0 if both adjacent spots of the target parking spot are occupied, 0.5 if only one of the adjacent spots is occupied and 1.0 if neither adjacent spot is occupied.
- *Likelihood of availability* (s_{avail}): likelihood that the user will find at least one available parking spot along the same road segment as the target parking spot at the time of arrival.

The system estimates the likelihood of availability based on a predefined statistical model. We assume that new vehicles arrive at available parking spots, following a Poisson process with the arrival rate λ being 0.05 vehicles per second. Let k denote the current number of available spots along the same road segment as the target parking spot. The probability that at least one of them remains open until the user vehicle arrives in the road segment can be calculated using the cumulative distribution function of the Poisson distribution as:

$$s_{avail} = \exp(-\lambda) \sum_{i=0}^{k-1} \frac{\lambda^i}{i!}. \quad (1)$$

Thus the parking search paths from each pair of options A and B are represented as 4-tuples $s^{(A)} = (s_{walk}^{(A)}, s_{drive}^{(A)}, s_{ease}^{(A)}, s_{avail}^{(A)})$ and $s^{(B)} = (s_{walk}^{(B)}, s_{drive}^{(B)}, s_{ease}^{(B)}, s_{avail}^{(B)})$, respectively.

We define a personalized parking selection model as a function that takes an 8-dimensional input vector $(s^{(A)}, s^{(B)})$ as input and then outputs a user's preferred search path (*i.e.*, A, B or neutral) under the given parking availability condition. Since different users have different parking preferences, this function should be tuned on a per-user basis. While there are a variety of machine learning frameworks that can be applied to the preference modeling, we use a simple decision tree algorithm in this initial feasibility study. Out of 50 responses from each survey participant, we used 40 of them to train the personalized decision tree and the remaining 10 responses for performance evaluation. We limit the depth of the decision tree to two, which we have found to give the most accurate results. Thus each user's decision on parking spot selection is modeled as up to two cascaded if-then-else rules, which are tailored to individuals.

4 PREDICTION PERFORMANCE

Figure 2 shows confusion matrices with (a) a generic parking spot preference model which is trained over all the responses from 84 survey participants and (b) personalized models that are trained on a per-user basis. The generic model results in many prediction errors with the subset accuracy being 0.49. This fits our intuition that users have various preferences on parking spot selection. In contrast, the personalized parking preference models achieve the subset accuracy of 0.73, which clearly shows the benefit of user-specific parameter fitting. It would be worth noting, though, that the prediction accuracy for the "neutral" category is lower than

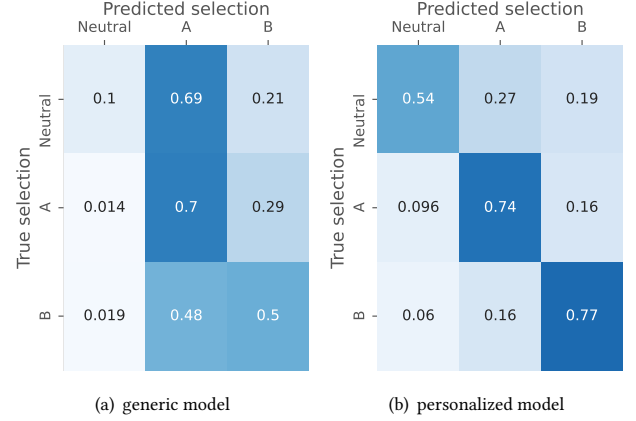


Figure 2: Confusion matrices

the cases that a user preferred one parking search path over the other. This implies that users' selection of parking spots may not be always consistent, especially when the given candidate spots are equally preferable.

5 CONCLUSION AND DISCUSSION

The results in the previous section show the feasibility of predicting user's parking spot selection based on the history of their decisions in the past. In order to improve generalizability and accuracy of the prediction models, it would be worth to investigate (i) other possible features to characterize candidate parking search paths, (ii) the impact of the size, type and layouts of parking lots on parking spot selection and (iii) other machine learning frameworks that have richer representation capabilities.

Another open challenge is to build such parking spot preference models from vehicle-generated sensor data (*e.g.*, GPS probes, crowd-sourced parking spot availability information, etc.). If a vehicle in parking search passes an available parking spot and is parked in another spot, the system may reasonably assume that the driver preferred the latter spot to the former. Analysis of the history of such parking search paths, combined with the spot availability information, would be a promising solution to automate the preference modeling.

Once the system builds the personalized preference model, the last step is to calculate a preferred parking search path that maximizes user satisfaction. This constitutes another important research problem to be investigated in our future work.

REFERENCES

- [1] Graham Cookson and Bob Pishue. 2017. The Impact of Parking Pain in the U.S., U.K. and Germany. <http://www2.inrix.com/research-parking-2017>.
- [2] Takamasa Higuchi and Kentaro Oguchi. 2020. Monitoring Live Parking Availability by Vision-based Vehicular Crowdsensing. In *Proc. IEEE GLOBECOM*. 1–5.
- [3] Suhas Mathur, Tong Jin, Nikhil Kasturirangan, Janani Chandrasekaran, Wenzhi Xue, Marco Gruteser, and Wade Trappe. 2010. ParkNet: Drive-by Sensing of Road-side Parking Statistics. In *Proc. ACM MobiSys*. 123–136.
- [4] San Francisco Municipal Transportation Agency. 2014. SFPark Pilot Project Evaluation Summary: A summary of the SFMTA's evaluation of the SFPark pilot project.